Generations of Computers

A generation marks a major change in the **technology used to build computers**. Each generation became smaller, faster, cheaper and more reliable than the one before.

First Generation (1940-1956): Vacuum Tubes

- **Technology:** vacuum tubes for circuits, magnetic drums for memory
- **Language:** machine language (0s and 1s)
- **Features:** huge in size (filled whole rooms), very expensive, produced a lot of heat, consumed much electricity, unreliable and slow
- **Input/output:** punched cards and printouts
- **Examples:** ENIAC, UNIVAC, EDVAC, IBM 701

Second Generation (1956-1963): Transistors

- **Technology:** transistors replaced vacuum tubes
- **Language:** assembly language, early high-level languages (FORTRAN, COBOL)
- **Features:** smaller, faster, cheaper and more reliable, less heat and power use
- **Memory:** magnetic core
- **Examples:** IBM 1401, IBM 7090, CDC 1604

Third Generation (1964-1971): Integrated Circuits (IC)

- **Technology:** many transistors placed on a single silicon chip (IC)
- **Language:** high-level languages (BASIC, Pascal, C)
- **Features:** much smaller and faster, more reliable, lower cost. Introduced keyboards, monitors and operating systems that supported multitasking.
- **Examples:** IBM 360 series, PDP-8, Honeywell 6000

Fourth Generation (1971-Present): Microprocessors

- **Technology:** VLSI (Very Large-Scale Integration), with thousands to millions of transistors on one chip. The first microprocessor was the **Intel 4004 (1971)**.
- **Language:** C, C++, Java, Python and other high-level languages
- **Features:** personal computers (PCs), GUI, mouse, networking and the Internet, laptops, portable and affordable
- **Examples:** IBM PC, Apple Macintosh, modern desktops and laptops

Fifth Generation (Present and Future): Artificial Intelligence

- **Technology:** ULSI (Ultra Large-Scale Integration), parallel processing, AI, machine learning, quantum computing research
- **Features:** voice recognition, natural language processing, robotics, cloud computing, self-learning systems
- **Examples:** AI assistants, smart systems, supercomputers

Comparison Table

Generation	Period	Technology	Size	Speed
First	1940-56	Vacuum tubes	Room-sized	Very slow
Second	1956-63	Transistors	Smaller	Faster
Third	1964-71	Integrated circuits	Small	Much faster
Fourth	1971-now	Microprocessors (VLSI)	Very small	Very fast
Fifth	Present/future	AI, ULSI	Compact	Extremely fast

SMPS (Switched Mode Power Supply)

What is SMPS?

The SMPS is the **power supply unit (PSU)** of a computer. It takes **AC (alternating current)** from the wall socket, usually 230V in India, and converts it into the **low-voltage DC (direct current)** that computer parts need, such as +3.3V, +5V and +12V.

It is called "switched mode" because it uses **high-speed electronic switching** (transistors turning on and off thousands of times per second) to regulate voltage efficiently.

Why Do We Need It?

Computer components work on low-voltage DC. Connecting them directly to AC mains would destroy them. The SMPS also keeps the voltage stable and protects the system.

Working Stages of SMPS

1. **Input rectifier and filter:** converts AC to rough DC and smooths it using capacitors.
2. **Switching circuit:** a transistor switch chops the DC at a very high frequency.
3. **Transformer:** a small, high-frequency transformer steps the voltage down.
4. **Output rectifier and filter:** converts the stepped-down signal back to smooth, stable DC.
5. **Feedback and control circuit:** monitors the output voltage and adjusts the switching to keep it constant.

Flow: AC input → Rectifier → Filter → Switching → Transformer → Rectifier → Filter → DC output

Main Parts of an SMPS

- **Fuse:** protects against overcurrent
- **Rectifier (diodes):** converts AC to DC
- **Capacitors:** filter and smooth the voltage
- **Transformer:** steps the voltage down
- **Switching transistors (MOSFET):** perform high-speed switching
- **Cooling fan:** removes heat
- **Connectors:** deliver power to components

Output Voltages and Their Use

Voltage	Used for
+12V	CPU, graphics card, fans, hard disk motors
+5V	USB ports, SSD/HDD logic, some motherboard circuits
+3.3V	RAM, chipset, M.2 and PCIe devices
-12V	Older serial ports (rarely used now)
+5V standby	Keeps the PC ready to start when the power button is pressed

Common SMPS Connectors

Connector	Powers
24-pin ATX	Main power to the motherboard
4/8-pin CPU (EPS)	The processor
PCIe 6/8-pin	Graphics cards
SATA power	Hard disks, SSDs, optical drives
Molex (4-pin)	Older drives and fans

Power Rating (Wattage)

SMPS capacity is measured in watts (W), for example 450W, 550W, 650W. A basic office PC needs about 300-450W, while a gaming PC with a powerful graphics card needs 650W or more.

80 PLUS certification (Bronze, Silver, Gold, Platinum, Titanium) shows how efficient the unit is. Higher means less energy wasted as heat.

Advantages of SMPS

- High efficiency (about 70-90% or more)
- Small, light and compact
- Less heat produced
- Works over a wide input voltage range
- Provides stable output

Disadvantages

- More complex circuit design
- Can produce electrical noise (interference)
- Repairs are harder than for older linear supplies

SMPS vs Linear Power Supply

Feature	**SMPS**	**Linear supply**
Efficiency	High (70-90%+)	Low (30-60%)
Size and weight	Small and light	Large and heavy
Heat	Less	More
Complexity	Complex	Simple
Noise	More	Less

Common SMPS Problems

- Computer does not turn on (dead SMPS)
- Random restarts or shutdowns
- Burning smell or loud fan noise
- Beeping or unstable power

Safety tip: never open an SMPS casually. Capacitors inside can hold a dangerous charge even after it is unplugged. Using a **UPS** or voltage stabilizer helps protect it from power fluctuations.